Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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Abstract

We develop a semiparametric downside-risk engine (GARCH scale, one pooled state-conditioned residual-quantile fit, an extreme-value tail, an optional conformal overlay) and a real-time misspecification score that predicts when it beats the parametric standard. Amortization: one fit serves hundreds of names and prices day-one listings from characteristics. Against standard benchmark models it reduces quantile loss (date-level DM 4.4–8.3), and aggregate date-clustered exception counts stay nominal at both FRTB levels through the 2008 crisis. The score concentrates the edge in its top decile (+2.7%, DM 6.5), survives a frozen-specification 2000–2013 holdout, Romano–Wolf control, and a point-in-time universe, and orders its magnitude everywhere.

Keywords: Value-at-Risk; Expected Shortfall; FRTB; quantile regression; extreme value theory; conformal prediction; misspecification; amortized estimation.

JEL classification: C14; C22; C52; C58; G17; G28.

1 Introduction

Two communities measure conditional downside risk. Econometrics builds parametric filters (GARCH and its leverage and skew extensions) whose conditional scale dynamics are excellent and whose innovation law is fixed by assumption.[1] Machine learning builds distribution-free conditional quantile estimators that assume nothing about the innovation law but must learn everything from data: gradient-boosted quantile trees, neural implicit quantile networks (IQN), and, most recently, generative Bayesian computation (GBC) in the sense of Polson and Sokolov (2023). Industry practice, meanwhile, is neither: under the Basel Fundamental Review of the Trading Book (FRTB), the regulatory objective is Expected Shortfall at the 97.5% level, the

[1] For readers outside econometrics: GARCH (generalized autoregressive conditional heteroskedasticity; Engle, 1982; Bollerslev, 1986) captures the most robust fact about markets, that volatile days cluster, by letting today's variance be a weighted average of yesterday's variance and yesterday's squared surprise (Eq. 2). "Leverage" extensions (GJR; Glosten et al., 1993) let down-moves raise volatility more than up-moves; "skew-t" versions allow an asymmetric heavy-tailed innovation. What all of them fix in advance is the *shape* of the standardized surprise; that assumption is what this paper's score monitors.

workhorse internal models are historical simulation (HS) and GARCH-filtered historical simulation (FHS), and backtesting proceeds through VaR exception counts at 99% and 97.5%. Any claim that a new method "beats the industry standard" must therefore beat HS and FHS on $ES_{97.5}$ under exception-test discipline; beating a vanilla GARCH on average loss is not enough.

One distinction comes first. When the misspecification score is quiet, on roughly ninety percent of asset-days, the engine we deploy and the parametric benchmark are statistically indistinguishable (Section 5 reports the widths): *no detectable loss on the accuracy yardstick.* On that yardstick the engine's worst cell in our tests is that bounded tie (the coverage shift's 2.5% joint-score concession, priced in Section 6, is the one measured exception), so holding the engine is cheap while it waits for the days on which it wins. What does lose on quiet days is a configuration we study only as a cautionary boundary and never deploy: a bare per-name neural quantile model, which a unit-matched GJR-GARCH-t beats by 4.5–6.2% at every horizon (DM t between +12 and +18)[2] That failure is what dictates the engine's hybrid design. The contribution is thus a *predictive theory of when flexible-shape methods win*, an engine that converts the theory into a risk model that beats the market-risk forecasting benchmarks banks commonly run on distributional accuracy while never doing worse than them on that yardstick, and a demonstration that the generative members of the family deliver capabilities no parametric point-forecasting model offers at all.

1.1 Thesis

A single measurable quantity — the local misspecification of the parametric conditional tail, operationalized as the excess kurtosis and asymmetry of recent GARCH-standardized residuals — governs when distribution-free quantile methods beat industry-standard parametric VaR/ES. Where that misspecification is high the deployed engine wins decisively and with statistical significance; where it is low the advantage shrinks toward zero — statistically indistinguishable from the benchmark in the design panel, a small positive in the point-in-time universe, and never a detectable loss. The score's robust content is the *ordering of magnitudes* (top decile roughly seven times the calm region), not a hard zero below a threshold. Everything else in the paper is either evidence for this claim or the engineering that makes the engine deployable in practice.

1.2 Contributions

1. **Amortization and the cold start** (Sections 3 and 7). The shape stage is a single amortized model trained across hundreds of names: one fit replaces per-asset estimation of the innovation shape (the conditional-scale filter remains the industry's standard per-name recursion), transfers to names the model has never seen, and beats own-history benchmarks at *every* listing age (the advantage largest for young listings). Its characteristics-only form,

[2] Throughout, DM denotes the Diebold and Mariano (1995) test: a t-statistic on the per-date mean loss difference of two forecasters (pinball loss unless stated), with a serial-correlation-robust standard error. The difference is oriented so that positive t favors the model reported as winning; the engine's headline comparisons use $d_t = \ell_t^{\text{benchmark}} - \ell_t^{\text{engine}}$. Headline tests are one-sided at the 5% level ($t > 1.645$); two-sided readings are noted where used.

which needs no scale filter and hence no history, prices assets with no return history at all, the one regime where no per-asset model exists.

2. **An FRTB-aligned risk engine** (Section 6). The residual-hybrid (GARCH conditional scale $\times$ state-conditioned nonparametric residual quantile) with an extreme-value left tail and an optional split-conformal coverage overlay: co-best with CAViaR[3] in the 90% Model Confidence Set[4], significantly better than GARCH-t/FHS/EWMA/HS/GJR-skew-t, exception counts consistent with nominal at both regulatory levels (Kupiec+Christoffersen[5] at 99% via the EVT tail; conformal recalibration at 97.5%); nominal on the aggregate date-clustered tests and through a 2008-crisis window rerun, with per-name pass rates disclosed as residual heterogeneity; the lowest joint $(\mathrm{VaR}, \mathrm{ES})$ FZ0 score at both levels on the full panel for the engine's accuracy layer, with the conformal shift's sharpness-for-coverage cost at 2.5% quantified rather than hidden (and removed overall by its adaptive form). A separate direct multi-day extension, not the residual-hybrid engine, shows the edge growing at the ten-day horizon in 2014–2024 and reversing out of era.

3. **The misspecification frontier** (Section 5). A live, per-asset, per-day score that predicts when nonparametric quantiles beat GARCH-t, with per-date Diebold–Mariano significance, validated across four universes: CRSP US equities (top-decile edge +2.71%, DM 6.54), FX (hyperinflation tier pooled DM 4.12), twenty-six country indices (developed→frontier gradient, corr 0.53), and forty-three cross-asset instruments (corr 0.43), with Korea-2026 as a sharpening negative control: price crashes are not residual misspecification.

4. **A theory of where the edge lives** (Section 3, proofs in Appendix A). A pinball regret identity showing that a forecaster's excess loss is quadratic in its quantile wedge, so any edge must concentrate where the wedge is largest, and a tail-wedge proposition showing that innovation-shape misspecification creates that wedge in the tail — together the motivation for looking where the empirical edge in fact appears.

Some neighboring questions are out of scope and claimed nowhere in this paper: posterior uncertainty bands on the reported $(\mathrm{VaR}, \mathrm{ES})$; likelihood-free amortized posteriors for parametric simulators; and the multivariate portfolio tail. This is a point-forecasting paper, and it is self-contained: every stage the deployed engine uses is developed, and every formal result it relies on proved, here.

Summary of results. Table 1 states every headline magnitude with its test statistic. Panel A is the presented engine's record: its worst cell on the accuracy yardstick is a *tie*, and its one

[3]CAViaR (Engle and Manganelli, 2004) models the quantile itself as an autoregressive process estimated on the pinball loss; the strongest non-nested rival in the battery.

[4]The MCS (Hansen et al., 2011) is the subset of models that cannot be statistically rejected as best at a given confidence level; "co-best" means surviving the elimination sequence.

[5]Kupiec (1995) tests whether the *number* of days losses exceed VaR matches the nominal rate; Christoffersen (1998) additionally tests that exceptions arrive *independently* rather than in clusters. Both are standard in regulatory backtesting.

scored concession, the coverage shift's 2.5% joint-score cost, is flagged in its own row rather than absorbed into an average. The ten-day line sits in its own block because it belongs to the direct multi-day extension, not the engine. Panel B is not the engine's record at all: it collects the boundary evidence, configurations we deliberately do *not* deploy and negative controls, that motivates the engine's design and delimits where the flexible-shape edge lives.

The asymmetry of the loss profile is the economic argument for adoption: when the frontier is quiet the engine ties or edges the parametric benchmark (calm-quintile edge ≈ 0, never negative), so holding it costs at most a rounding error while it collects +2.7% in the decile where risk models actually fail and 12–14% where the parametric assumption breaks entirely. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level; the joint FZ0 score of Section 6 carries the economic comparison, and no capital-release arithmetic is attempted.

Throughout, the exposition follows three reporting conventions fixed in advance. First, the strongest tabular estimator[6] in the family is gradient-boosted trees, not the neural network; we present GBC as the framework, trees as the strongest estimator, and the tail-aware neural IQN as the canonical generative realization, competitive after tuning and uniquely able to sample. Second, residual kurtosis is necessary but not sufficient for a nonparametric win (carbon and corn are high-kurtosis ties; Baltic freight wins at moderate kurtosis through limit-move dynamics); the frontier is a strong regularity with counterexamples, and we show them. Third, every negative result that shaped the design — the retired same-universe artifact, the electricity log-return artifact, the failed hierarchical blends, the failed learned gate, the three multivariate losses — is reported in the text.

2 Background and industry benchmark

2.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail) and the return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$; the regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 3.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at $\mathrm{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket) while backtesting remains VaR-exception based: more than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint: accuracy on $\mathrm{ES}_{97.5}$ and pinball, *and* unconditional (Kupiec, 1995) (the Kupiec test: are there the right number of VaR breaches?) plus independence (Christoffersen, 1998) (the Christoffersen test: are breaches independent, rather than clustered in time?) exception tests at both levels, with significance assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) (a t-test on the difference in loss between two forecasts) and the

[6] "Tabular" data: observations that are rows of engineered features (here, the state vector s_t) rather than raw sequences or images — the setting where tree ensembles remain the strongest general-purpose learners.

Model Confidence Set (Hansen et al., 2011) (the set of models statistically indistinguishable from the best one).

2.2 The parametric family, the simulation family, and CAViaR

Our battery spans the models banks actually run: historical simulation; EWMA/RiskMetrics (Zumbach, 2007) (an exponentially weighted moving average of squared returns, giving recent days more weight); GARCH(1,1)-t (Bollerslev, 1986); GJR-GARCH-skew-t (adding leverage and skewness; Glosten et al., 1993); GARCH-FHS (parametric scale, empirical standardized residuals — the volatile-regime best practice; Barone-Adesi et al., 1999); and SAV-CAViaR (Engle and Manganelli, 2004), included after an anti-strawman audit showed its omission flattered our model. CAViaR (conditional autoregressive Value-at-Risk) skips modeling the distribution entirely and lets the quantile itself follow a recursion estimated on the pinball loss; the SAV (symmetric absolute value) specification is $q_t(\tau) = \beta_0 + \beta_1 \, q_{t-1}(\tau) + \beta_2 \, |r_{t-1}|$ — today's VaR is yesterday's VaR adjusted by the size (not sign) of yesterday's move, making it the semiparametric analogue of a GARCH recursion written directly on the quantile.

2.3 Distribution-free conditional quantiles and GBC

The closest antecedent in this journal's own pages is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names — evidence of a shared volatility mechanism that our amortized design also exploits. The differences delimit this paper's contribution. Their target is a point forecast of realized variance; ours is the full conditional return distribution and its regulatory tail, held to calibration, exception-test, and capital discipline. Their question is whether machine learning wins on average; ours is *when* — and our answer, that on roughly ninety percent of asset-days it does not, reads as a caution against extrapolating average ML superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at every listing age. And we reach the opposite estimator verdict on tabular state (trees over networks), a divergence we attribute to the different target and feature set rather than a contradiction.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (Koenker and Bassett, 1978): gradient-boosted quantile trees (GBM) and the implicit quantile network of Dabney et al. (2018) (IQN: the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). Casting GBC as ERM conditional-quantile regression on observed pairs removes its forward-simulator requirement and extends it to observational time series. The deployed shape model is the trees; the IQN is retained as a benchmark, and its generative reading (draw $\tau \sim U(0,1)$, output $Q_\theta(\tau \mid x)$) is a capability we note without using, since sampling and posterior uncertainty are outside this paper's scope. Generalized Bayes enters through the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\}\pi(\theta)$, a posterior built from a loss in place of a likelihood, targeting the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016); distribution-free finite-sample (exchangeability-based) guarantees enter through split-conformal and adaptive conformal inference (recalibrating predicted intervals on held-out data so that coverage holds without a parametric distributional model, under exchangeability) (Gibbs and Candès, 2021).

3 Methods

This section is written to be self-contained: a reader from any one of econometrics, machine learning, or risk management should be able to reimplement every estimator from the formulas and algorithm boxes below. (For a finance-journal submission the algorithm boxes and the proof of Proposition 1 can migrate to an appendix unchanged; they are self-contained.) A cross-field glossary of acronyms and jargon is provided in the Online Appendix.

3.1 Notation and objects

- r_t — return of an asset on day t; $\mathcal{F}_{t-1}$ — information available before t.
- $Q_t(\tau)$ — the conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$, formally $Q_t(\tau) = \inf\{q : \Pr(r_t \leq q \mid \mathcal{F}_{t-1}) \geq \tau\}$: the number the return falls below with probability τ. One sign convention is used throughout: the paper works in *return space*, $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1} \int_0^\tau Q_t(u)\, du \leq q_\tau$ — the convention of the FZ scoring function ($v, e < 0$) and of every table (ES values such as -6.37 are return-space conditional means). The regulatory *loss-space* numbers are their negatives, $\mathrm{VaR}_\tau^{\mathrm{loss}} = -q_\tau$ and $\mathrm{ES}_\tau^{\mathrm{loss}} = -e_\tau$, used only when quoting FRTB levels; the two are never mixed in one expression.
- $z_t = (r_t - \mu_t)/\sigma_t$ — the *standardized residual*: the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law.
- s_t — the state vector (trailing realized volatility, residual-shape scores, lags) on which non-parametric quantiles condition.
- $\tau \sim \lambda$ — a quantile level drawn from a base distribution λ; $H_\phi(s, \tau)$ — a learned generator mapping (state, level) to a quantile, in the sense of Polson and Sokolov (2023): $z \stackrel{D}{=} H(s, \tau)$, $\tau \sim U(0, 1)$, is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u < 0\}), \qquad u = z - q, \tag{1}$$

whose expectation is uniquely minimized by the true conditional quantile (Koenker and Bassett, 1978): differentiating $\mathbb{E}\,\rho_\tau(z - q)$ in q and setting it to zero gives $\Pr(z < q) = \tau$, so the loss *elicits* the quantile the way squared error elicits the mean — under-prediction is charged τ per unit, over-prediction $1-\tau$, and the charges balance exactly at the τ-quantile. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below), and quantile functions carry a natural L_1 geometry, $W_1(F, G) = \int_0^1 |F^{-1}(\tau) - G^{-1}(\tau)|\, d\tau$. Both facts are density-free, which is what permits every estimator below to dispense with a likelihood.

Why this loss and not another. The choices among standard alternatives are deliberate. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\, d\tau$ (Gneiting and Raftery, 2007); scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail — CRPS would average those differences away against a body where all competitors agree. We therefore score pinball *at the levels that are decision-relevant*, and use tail-weighted τ-sampling in training for the same reason. (b) Expected Shortfall is not elicitable on its own — no loss function is minimized by the true ES alone — but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016); the FZ-scored re-evaluation of the battery is a pre-committed extension (Section 8). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them.

3.2 The final model: the residual-hybrid engine

The deployable model of this paper is *not* a bare neural network; it is a four-stage composition, each stage keeping what an industry model gets right. The design principle that survives every experiment is a division of labor: *let the parametric filter do what it is good at (conditional scale), and spend the nonparametric capacity only on what it is bad at (the conditional shape of the innovation law).* Intuitively: GARCH answers "how big is a typical move today?" extremely well, and badly answers "given a typical move size, how likely is a move five times that, and is down likelier than up?" The engine keeps GARCH's answer to the first question and learns the second from data. Each finishing stage has the same one-line logic: the EVT tail says *beyond the data, extrapolate with the law extreme-value theory motivates for high-threshold exceedances, instead of trusting a fitted model outside its training range*; the conformal stage says *measure your own miss rate on data you have not touched, and shift the curve by that measured miss* — calibration by bookkeeping, exact under exchangeability (Section 3.2).

Why one should expect this to work before seeing data. The architecture is not an empirical accident; each piece is where prior reasoning puts it. Conditional *scale* is the strongest, best-understood signal in daily returns (volatility clustering is the most robust stylized fact in the field), and it is identified by a handful of parameters — so a parametric filter estimates it with minimal variance, and a nonparametric model that re-learns it merely spends data on what four parameters already buy. Conditional *shape*, by contrast, is where the parametric family is constrained by assumption and where per-asset data is too scarce to fit flexibly — tail events are rare by definition — so that is where nonparametric capacity earns its keep, and only if it can borrow strength across assets. This is classical bias–variance division of labor: parametric where the structure is known, flexible where it is not, which is why the hybrid should (and does) dominate both endpoints. Pooling residuals across names is in turn the standard hierarchical-shrinkage argument: once name-specific scale is divided out, standardized surprises are approximately draws from a common state-conditional law, and estimating a shared object from the whole cross-section improves on any single name's estimate whenever units are scarce individually and similar collectively — the same logic that makes partial pooling dominate in hierarchical Bayes. The EVT stage is backed by a limit theorem: exceedances over a high threshold converge to the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), so tail extrapolation by GPD is the principled choice for *any* body model. And the conformal stage's guarantee is Proposition 1, not an empirical observation. The experiments in this paper are therefore tests of a mechanism argued in advance, not a specification search rewarded after the fact.

Stage 1 (parametric scale). Fit GARCH-t,

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by quasi-maximum likelihood on a trailing window, and keep only $\widehat{\sigma}_t$ and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Pickands, 1975; McNeil and Frey, 2000); the ES formula below requires $\xi < 1$, a condition every fitted tail in this paper satisfies by a wide margin and the code guards explicitly.[7] On losses $X = -\widehat{z}$, fix a threshold u at the empirical $(1 - p_0)$ loss quantile with $p_0 = 0.025$, fit the generalized Pareto distribution (GPD) $G_{\xi,\beta}(x) = 1 - (1 + \xi x/\beta)^{-1/\xi}$ — shape ξ sets how heavy the tail is, scale β how wide — to exceedances $X - u > 0$ strictly on past data, and set, for $\tau \le p_0$,

$$\widehat{q}^{\,\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\,\mathrm{EVT}}(p_0) = -u$ (and reduces to $-[u + \beta\log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit; requires $\beta > 0$ and $1 + \xi(x - u)/\beta > 0$). Since u is the *unconditional*

[7] Extreme-value theory has two classical formulations, and the distinction matters for reading the literature. The *block-maxima* approach models the maximum of each block (e.g., yearly worst loss), which converges to the generalized extreme value (GEV) family — Gumbel, Fréchet, or Weibull type depending on the tail. The *peaks-over-threshold* (POT) approach, used everywhere in this paper, models all exceedances over a high threshold, which converge to the generalized Pareto distribution (GPD) (Balkema and de Haan, 1974; Pickands, 1975). POT uses the data more efficiently (every large loss contributes, not just one per block), which is why it is the standard in risk management (McNeil and Frey, 2000; Embrechts et al., 1997). The GPD's shape parameter ξ corresponds to the GEV type: $\xi > 0$ is the heavy-tailed (Fréchet) case relevant for financial losses.

empirical loss threshold while the Stage-2 body quantile at p_0 is the *state-conditional* $\widetilde{q}(p_0 \mid s)$, the splice is exactly continuous only when the threshold is anchored at the body's own p_0-quantile, $u = -\widetilde{q}(p_0 \mid s)$; otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join and the two pieces are spliced to agree in practice by construction. The tail is applied selectively where the raw tail under-covers (it *hurts* some volatility-index series and is not applied universally).

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$ and shift the curve per level by the empirical correction c_τ chosen as the $\lceil (n+1)\tau \rceil$-th order statistic of $\{e_i\}$: $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Romano et al., 2019). This distribution-free layer repairs the 97.5% level that every model in the battery otherwise fails.

Proposition 1 (split-conformal validity under exchangeability)**.** *If the calibration and test residuals $e_1, \ldots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \leq n$ the shifted quantile satisfies*

$$\tau \;\leq\; \Pr\left(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\right) \;=\; \frac{\lceil (n+1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \ldots, e_{n+1}\}$ is uniform on $\{1, \ldots, n+1\}$; the event $z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \leq e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \leq \lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument, the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1), and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$, are in Appendix A. The guarantee is marginal, not conditional on s_{n+1}. □

The proposition's scope is narrower than its use. In deployment the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable; the proposition therefore *motivates* the construction, while the evidence that it works here is empirical — the exception-test repairs of Section 6 (the conformal stage is the only reason any entrant passes Kupiec at 97.5%). Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018); we prefer the simple split form plus an empirical audit. One further implementation gap is disclosed rather than hidden: the first-stage GARCH is fit on the full pre-test history, calibration window included, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer contests never read the calibration split and share their filter with the benchmark (identical information sets); the overlay's shift does use it, so the audit was rerun end-to-end strict, filter estimation stopped at the calibration boundary. Against the matched-information benchmark the advantage is intact (DM 5.1 and 5.4 at the two levels); the strict engine's gap to the full-window benchmark (−0.6, −2.5) matches that benchmark's own margin over its short-window twin (2.8, 3.8) — a quarter less estimation data, not leakage (replication artifacts: `job_fz_strict_calibration.py` → `fz_strict_calibration_results.json`). The static shift widens (per-name pass 34% against 48%), consistent with its disclosed conservatism. Run under this same strict split, the adaptive update (Gibbs and Candès, 2021) walks the shift from −0.51 to −0.20, lifts the per-name pass rate to 57%, and shrinks the joint-score gap to GARCH-t from DM −3.2 (static) to −1.5, not significant at the two-sided 5% level: the adaptive form is established as the remedy under the same information architecture. The engine's annual-refit schedule sits between the static and adaptive poles, and the exception tests decide between them.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) \;=\; \widehat{\sigma}_t \cdot \mathrm{R}\Big[\; \min\{\widetilde{q}_\phi(\tau \mid s_t),\, \widehat{q}^{\,\mathrm{EVT}}(\tau)\}\, \mathbb{I}\{\tau \le p_0\} \;+\; \widetilde{q}_\phi(\tau \mid s_t)\, \mathbb{I}\{\tau > p_0\}\Big], \tag{5}$$

where $\mathrm{R}[\cdot]$ is the monotone rearrangement across levels. The tail forecast is the pointwise more conservative branch (a feature, not a guard: the body branch is the minimum on 38% of tail nodes), and both risk measures come from this one curve. Recomputing ES_α as the exact integral of (5) rather than the GPD closed form leaves every comparison in place (at 2.5%, FZ0 1.849 against 1.851, the integral marginally better, DM 5.3; engine-over-GARCH DM 4.8 at both levels), and the splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$: DM 4.6–4.8).

Industry models as special cases. Equation (5) nests the standard toolkit. Historical simulation: $\widehat{\sigma}_t \equiv 1$ and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS): q the unconditional empirical quantile of past *residuals* — Stage 2 with the state dependence deleted. GARCH-t: $q = t_\nu^{-1}$, a fixed parametric shape. The hybrid is therefore "FHS and above": whatever FHS captures, (3) can represent, plus conditional shape. (CAViaR sits outside the nesting — it autoregresses on the quantile directly, with no scale/shape split — and is the one rival the engine only ties, Section 6.)

Algorithm 1 (residual-hybrid engine). (i) Fit (2) per asset on a trailing window; store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names; train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances; splice at p_0. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5). In *production* all stages refit on an annual walk-forward schedule; the paper's *evaluated* protocol freezes every stage once per split (the more conservative test — no stage ever adapts to the test era), and Section 4.1 verifies the frontier separately under the true annual-refit schedule.

3.3 Estimating the shape: trees for accuracy, networks for generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3), and the paper is deliberate about which to use when (Table 2).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise, each new tree h_m fit to the negative gradient of the pinball loss (τ_k where the current model under-predicts, $\tau_k - 1$ where it over-predicts); the fitted grid is made non-crossing by monotone rearrangement — sorting the values $\{q(\tau_1|s), \ldots, q(\tau_K|s)\}$ in τ, which never increases the loss (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest *point* estimator — about 0.75% better pinball than the tuned network — consistent with the broader tabular-learning literature.

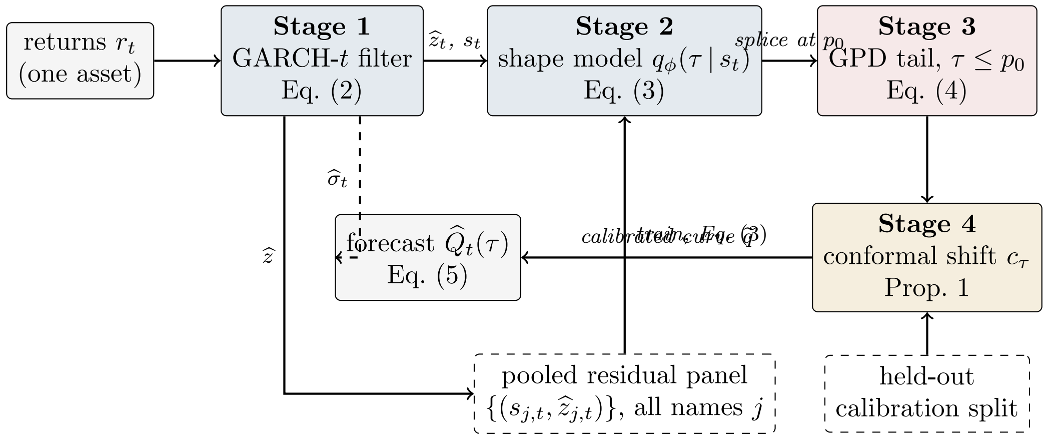

Figure 1: The residual-hybrid engine as a pipeline. Solid arrows carry data; the dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve; dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal overlay). In production every stage refits annually using only past data; the evaluated protocol freezes each stage once per split, and the annual-refit rerun is reported separately.

The neural implicit quantile network (IQN; the GBC estimator). A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s,\tau) \;=\; g\big(\psi(s)\circ\varphi(\tau)\big), \qquad \varphi_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=0}^{n_c-1}\cos(\pi i\tau)\, w_{ij} + b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Its modifications, each forced by a documented failure, make it competitive on financial data: tail-aware τ-sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, oversampling the extremes that uniform sampling starves (the raw net's 99% breach rate was 3.2%); monotone rearrangement of the τ-curve; and the conformal layer of Section 3.2. The network's earning is not point accuracy but *generation*: with $\tau \sim U(0,1)$, $z^* = H_\phi(s,\tau)$ is an exact sampler of the fitted conditional law (the inverse-CDF representation) — a capability no tree ensemble or GARCH filter provides.

How training works, concretely. The objective is the empirical version of (3): mini-batch stochastic gradient descent on $\frac{1}{|B|}\sum_{(j,t)\in B}\rho_{\tau_{jt}}\big(\widehat{z}_{j,t} - H_\phi(s_{j,t}, \tau_{jt})\big)$, where each example in each batch receives a *fresh* draw $\tau_{jt} \sim \lambda$ every epoch — over training the network sees every observation at many random levels, which is what lets a single set of weights carry the entire quantile curve. Early stopping is on held-out pinball loss; monotone rearrangement and the conformal shift are applied *after* training as post-processing; and the whole pipeline (filter, shape model, tail, shifts) is refit on the annual walk-forward schedule using only past data at every step. Exact layer sizes, learning rates, and seeds are recorded in the project's run configurations and reported with the code release rather than inline.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names; in generative-Bayes terms, H is an amortized map evaluated at any new (s,τ) without per-asset refitting. The intuition for why this works, and why it works at cold-start in particular, is actuarial. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns one *universal mapping from observable state to residual shape* ("assets whose trailing volatility looks like this, with characteristics like that, produce surprises shaped like so"), estimated from millions of asset-days of *other* names. A newly listed asset is not a new problem: its first ticks and characteristics place it at a point of state space already densely populated by other names' history, much as an insurer prices a new driver from the risk table built on millions of existing drivers rather than from the new driver's empty claims file. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically. The same fact explains why the explicit own-history blends fail: the state vector already carries the name's own recent information, so bolting it on again adds only noise. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's *own recent dynamics* (realized vol above all; removing it costs +1.0%, removing all cross-sectional characteristics only +0.5%), with characteristics mattering at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update, both *hurt* at every listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 (benchmark tier) on the M5 competition data, outside finance altogether.

Practitioner's selection rule. The two estimators minimize the same loss over different index sets — the trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level; the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum — and the choice follows from what the application consumes. If the desk needs *numbers at fixed regulatory levels* (VaR/ES reporting, limits, capital), use the trees: best point accuracy, no GPU, no sampling needed. If the application consumes *draws* — scenario generation, portfolio aggregation by simulation — use the network: it learns one continuous conditional quantile map, so draws come by inverse transform with no grid interpolation, at a measured cost of $\sim$0.75% in point accuracy (a GARCH-t or FHS model also simulates, from its fixed innovation law; a tree grid requires monotone interpolation first). If both are needed, run both from the same pooled residual panel; they share every upstream and downstream stage of (5), so the swap is one component, not a new pipeline.

Algorithm 2 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t. (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z} - H_\phi(s, \tau))$ [network], or boost per-τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset — including one with no return history, using characteristics-only s — evaluate $H_\phi(s_t, \cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0,1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

3.4 Formal results: why shape misspecification must show up in the loss

The paper's organizing claim — the nonparametric edge lives where the fixed innovation shape is locally wrong — is an empirical finding with a structural motivation in the pinball loss. The results below are deliberately modest: a shape violation creates a quantile wedge, and regret is quadratic in that wedge — motivation for looking where trailing shape diagnostics are extreme, not proof that mk_{63} estimates the wedge; that link is established empirically.

Lemma 1 (pinball regret identity)**.** *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ (so S is finite) and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(u) - \tau\big)\, du \;\geq\; 0, \tag{7}$$

with equality iff $F \equiv \tau$ on the interval between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof sketch. The difference quotients of $q \mapsto \rho_\tau(Z - q)$ are dominated by the Lipschitz constant $\max(\tau, 1 - \tau)$, so differentiation under the expectation is justified and $S'(q) = F(q^-) - \tau$ (a.e. equal to $F(q) - \tau$); integrating from q_F to q gives (7) *for any* F, atoms forming a null set. The integrand $F(u) - \tau$ has the sign of $u - q_F$ (by definition of q_F), so the integral is ≥ 0 whichever side q lies; (8) then follows from $F(q_F) = \tau$ (supplied by the density hypothesis) and $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$ for almost every u between q_F and q. Full details, including both orientations of the integral, are in Appendix A. □

Lemma 1 says the extra loss a misspecified forecaster pays is *quadratic in its quantile error*. The frontier claim then reduces to: does shape misspecification create a quantile wedge at the risk levels that matter? For the paper's canonical case — a fixed Student-t shape when the true local shape is fatter — it must:

Proposition 2 (tail wedge under tail-shape misspecification)**.** *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. There exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2)$ such that for all $\tau < \tau^*$ and $\nu_1 < \nu_2$, $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$: the fatter-tailed law has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}\,(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ at every level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof sketch. The unit-variance t_ν ($\nu > 2$, so the variance is finite and the normalization is well defined) has a regularly varying left tail: $\Pr(Z < -x) = c_\nu\, x^{-\nu}(1 + o(1))$ as $x \to \infty$ with $c_\nu > 0$; unit-variance rescaling alters c_ν but not the index ν. Inverting a regularly varying tail (Embrechts et al., 1997) gives $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$. For $\nu_1 < \nu_2$ the exponent $1/\nu_1 > 1/\nu_2$ makes $|Q_{\nu_1}(\tau)|/|Q_{\nu_2}(\tau)| \to \infty$, so the strict ordering and the diverging wedge hold for all τ below some $\delta > 0$; taking $\tau^* = \min(\delta, 1/2)$ places τ^* in $(0, 1/2)$ as claimed. (The proposition asserts *existence* of such a τ^*, not the location of the finite-sample crossover; the latter is $\tau_c \approx 0.0179$ for the t_3-vs-normal case of Remark 1.) The regret bound is (8); full details in Appendix A. $\square$

Remark 1 (the crossover is real and matters). Because unit-variance fat-tailed laws put *more* mass near the center as well as in the far tail, the quantile ordering of Proposition 2 reverses at moderate levels: numerically, the unit-variance t_3 has a *less* extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$ (solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root), so the deepest deployable level $\tau = 0.01$ sits *below* the crossover (fat tail strictly more extreme) while $\tau = 0.025$ sits just *above* it (at 2.5% the unit-variance t_3 quantile is -1.837, still less extreme than the normal's -1.960). Two of the paper's design choices are direct consequences. First, this is why CRPS-style whole-distribution scoring can rank a fat-tail-aware model *below* a thin-tailed one even when the tail is the decision-relevant region (Section 3.1): body and far tail disagree in sign. Second, it is why the deepest deployable level straddles the crossover rather than sitting cleanly beyond it, and why the score of Section 3.7 is used as a *rank* rather than a moment estimate.

Lemma 2 (validity of the generative sampler). *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof sketch. For any x, left-continuity and monotonicity make $\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ an interval closed at its top, so $\{\widetilde{q}(\tau \mid s) \le x\} = \{\tau \le G(x)\}$ with $G(x) = \sup\{t : \widetilde{q}(t \mid s) \le x\}$. Since $\tau \sim U(0,1)$, $\Pr(Z^* \le x) = \Pr(\tau \le G(x)) = G(x)$, i.e. Z^* has CDF G. As G is the generalized inverse of $\widetilde{q}$ and $\widetilde{q}$ is nondecreasing and left-continuous, the generalized inverse of G returns $\widetilde{q}$ exactly (the standard inverse-of-inverse duality), so $\widetilde{q}(\cdot \mid s)$ is the quantile function of Z^*. Full statement in Appendix A. $\square$

Remark 2 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS; empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 6 test whether that dominance survives estimation noise. It does; the converse is that the same argument does *not* apply to CAViaR, which is why CAViaR is the one rival the engine only ties.

3.5 A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth (run once, seed fixed) checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the score: unit-variance Student-t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). First, the Lemma 1 identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 promises. Third, and most instructive, the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small; static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_ν law has infinite estimator variance for $\nu \leq 8$, so a moment-based score estimate is at its noisiest just when tails are fat, and sixty-three observations cannot reliably rank regimes that differ only in a static ν. In the empirical sorts this noise works *against* the frontier, not for it: misclassification across bucket boundaries attenuates measured top-bucket edges, so the reported concentration is if anything understated; the jump and asymmetry signals, whose top deciles independently survive familywise control, corroborate the ordering, and a tail-index (Hill-type) variant of the score is an obvious robustness extension. For the empirical sections this means the real-data top-decile edge of Section 5 cannot be an artifact of stationary fat tails being averaged over (that configuration is the one the simulation shows a fixed shape handles) and is consistent with *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 negative control reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

3.6 What is different from Polson–Sokolov GBC

Because Polson and Sokolov (2023) is this paper's methodological benchmark, we state the deltas explicitly; each is an innovation forced by the downside-risk target, and each is defined mathematically above. (1) *Base measure*: GBC trains with $\tau \sim U(0, 1)$; we train with the tail-weighted mixture $\lambda = \frac{1}{2}U(0, 1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$, because uniform sampling starves the extreme levels that downside risk is about (the uniform-trained net's 99% breach rate was 3.2%). (2) *Input space*: GBC maps raw data to posteriors; we first pass returns through the parametric filter (2) and model *standardized residuals*: the scale dynamics are the component the industry model already estimates well. (3) *Beyond-data tail*: no generative regression can know about quantiles beyond its training support; we splice the GPD tail (4) at p_0, replacing extrapolation by the network with extrapolation by extreme-value theory. (4) *Finite-sample guarantee*: GBC's output carries no coverage guarantee; the conformal stage adds an exchangeability-based one (Proposition 1), whose deployment status (motivation under time ordering, judged by the exception tests) Section 3.2 states. (5) *Amortization target*: GBC amortizes across *simulations of one model*; we amortize across the real cross-section of hundreds of assets, which is what creates the cold-start capability and the transfer results of Section 3.3. (6) *Scope discipline*: GBC is a computational method; this paper adds the empirical question GBC does not ask, *when* such machinery beats the parametric filter at all, and answers it with the score below.

3.7 The misspecification score

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape. Static heavy tails do not violate this — ν absorbs them. What violates it is the shape being *locally* wrong: recent residuals far fatter-tailed or more asymmetric than any fixed t. We therefore score, per asset-day, over the trailing window $W_t = \{t-63, \ldots, t-1\}$ of $n_w = 63$ residuals—the window *ends the day before the forecast target*, so every signal is lagged at least one day relative to the loss it is paired with, and the frontier sorts of Section 5 pair S_{t-1} with the day-t loss (this alignment is enforced in the estimation code by construction) with window mean $\bar{z}$ and standard deviation s_z:

$$\mathrm{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \mathrm{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

Terminology is fixed here once: *skewness* is the signed third moment sk_{63}; *asymmetry* always means its absolute value $|\mathrm{sk}_{63}|$ (departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign); everywhere the text says "asymmetry," the statistic computed is $|\mathrm{sk}_{63}|$. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-4,\ldots,t\}} |\widehat{z}_i|$. The frontier score is the decile rank of these within the trailing cross-section. All three are causal (trailing-window) and computable on any asset with a fitted GARCH — which is what makes the frontier a deployable monitor rather than an ex-post classification. Informally we will call this the misspecification *thermometer*: it reads the local temperature of the parametric assumption, and Table 2 is the treatment chart.

Why these statistics. The parametric model's one fragile assumption is that the standardized residual's *shape* is a fixed t_ν; its first two conditional moments are already absorbed by μ_t, σ_t. The lowest-order ways a fixed symmetric law can be locally wrong are precisely the third and fourth standardized moments: sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies (an exact t_ν sample has known positive excess kurtosis; the score is used as a *rank*, so only its cross-sectional and temporal variation matters, not its t_ν baseline). J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent — and Section 5 shows the realized edge concentrates there. Note the logical status of this choice: after μ_t and σ_t are fitted, the fixed innovation shape is the parametric model's *only* remaining free assumption. A meter that measures the local violation of that one assumption is therefore not one candidate signal among many — it is the residual degree of freedom itself, which is why one should expect it, a priori, to be the axis along which flexible-shape models win.

Algorithm 4 (the misspecification meter, as deployed). (i) Fit (2) on a trailing window; store $\widehat{z}_{t-62}, \ldots, \widehat{z}_t$. (ii) Compute mk_{63}, $|\mathrm{sk}_{63}|$, J_5 by (9). (iii) Convert each to a decile rank against the trailing pooled panel (same asset class); the score is the rank of the maximum. (iv) Act by Table 2: top decile $\Rightarrow$ the parametric model's average pinball loss runs ∼2–3% higher within this score region (a conditional average over the region, not a per-name point estimate) — flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 5.3). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

4 Data and evaluation protocol

Equity panels come from CRSP daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations; every modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: common shares (share codes 10 and 11) with at least 3,000 observations in the window, ranked by average market capitalization, top 200. Returns are in percent and include CRSP delisting returns where present; the one delisting event in the design panel moves headline results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data. Terminal access ended in mid-2026, so those exhibits are frozen at the version reported here and all continuing work runs on WRDS alone.

Licensed rows never leave the machines they were pulled to. The public repository carries code, configuration, and derived statistics; the panels rebuild from the queries above for any licensed subscriber. Table 3 gives the sample flow.

Splits are chronological within each name. The first 60% of a name's history is the estimation window and the last 40% is test; the final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH-t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on pooled estimation-window residuals, the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule's survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the conventions note that opens Section 5: mean pinball loss across eleven quantile levels for distributional accuracy, the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019) for joint (VaR, ES) pairs, per-date Diebold–Mariano statistics with a Newey–West variance for loss comparisons, the Model Confidence Set for multi-model fields, and Kupiec, Christoffersen, and date-clustered breach tests at the two regulatory levels for exception behavior.

"Frozen-specification" has a specific meaning in this paper, and we deliberately avoid the stronger word *registered*: no third-party registry or timestamp exists for these runs, so the claim is diffability, not notarization. A run is frozen-specification when the full specification — signals, features, hyperparameters, split rule, and test statistics — is fixed and the predictions are written down before the data are pulled. The holdout pipeline is line-for-line the design-era pipeline — both files are public in the replication repository — so a reader can diff the two and verify that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 5 ran under this rule. Frozen-specification runs that remain open are listed in Section 8.

4.1 Common shocks, calendar overlap, and the pooled learner

The design panels are split per name, 60/40 in each name's own time index. In an unbalanced panel such a rule can place one name's training window on another name's test dates, so a pooled learner could meet a systemic event in training that a per-name model would only meet out of sample — a leak that needs no individual feature to look ahead. We concede the point in principle: a per-asset split cannot by itself support a causal reading of a pooled edge. Here the channel is nearly degenerate (near-complete histories put split points on almost the same date), but nearly is not zero, and the question deserves an answer by construction.

The construction is a strict calendar-time split: every stage — each GARCH fit, the pooled learner, the GPD tail — is re-estimated on data strictly before 2020-01-01 and tested from that date for every name simultaneously, so no training window shares a single calendar day with any test window. The frontier survives: top decile +2.47%, per-date DM 6.22 over 1,258 dates (per-name split: +2.7%, DM 6.5); overall +0.71% (DM 1.87). The bottom decile is elevated here (+1.3%; a 2020–22 test era leaves few calm cells) but the ordering is intact — the top decile roughly doubles every other cell. The concentration is not a calendar-overlap artifact. A second cut targets the harder era: inside the 2000–2013 holdout, everything is re-estimated strictly before 2007-01-01 and tested on 2007–2013, so the global financial crisis is entirely absent from training for every name. The pooled learner still wins overall (+0.29%, per-date DM 3.08 over 1,762 dates) and the within-sample decile profile keeps its shape (top three deciles +0.45–+1.04% against +0.06–+0.09% in the bottom three); the frozen-threshold top cell of Section 5 is too thin in that era to test alone (3.4% occupancy, +0.85%, DM 0.7). A model that never saw a systemic crisis in training carrying a significant edge through the crisis it never saw is the opposite of what the leakage story predicts.

A last construction tests the schedule the paper recommends: the evaluated protocol freezes every stage once per split, so the score could in principle measure parameter staleness rather than shape. Under the true annual walk-forward — per-name GARCH refit on expanding data at each January-1 cutoff 2020–2024, residuals and score rebuilt under the refit parameters, pooled learner retrained — the top decile carries +2.47% at per-date DM 6.05 (lags 5–44: 6.6–5.4), indistinguishable from the frozen-fit frontier, while the overall edge compresses to +0.32% (DM 0.8): annual refits help the benchmark everywhere *except* where the score says shape is the problem. The frontier measures shape, not staleness.

4.2 Cross-sectional dependence, family-wise error, and the universe rule

Dependence, multiplicity, and survivorship receive the same treatment. *Dependence:* every headline test collapses the cross-section to one number per date before any variance is estimated (Section 5), so common-date dependence enters through the date series; bootstrap procedures resample dates, not asset-days. *Multiplicity:* the frontier grid spans three signals and ten deciles, and per-cell p-values ignore the thirty hypotheses' joint draw. We therefore report Romano–Wolf step-down family-wise-error control over the full 3×10 grid: per-date mean edges per cell, a stationary bootstrap over dates (mean block 10 days, B = 2,000; block 20 as sensitivity) that preserves cross-hypothesis dependence, and one-sided step-down adjusted p-values. Nine of the thirty cells survive 5% family-wise control under both block lengths, and they are the frontier's own: the top deciles of all three signals (mk_{63} $t = 9.4$, skew_{63} $t = 9.2$, jump_5 $t = 5.1$, adjusted $p < 0.001$), the ninth deciles of mk_{63} (adjusted $p = 0.022$) and skew_{63}, skew_{63}'s seventh, and jump_5's U-shaped far cells. Every mid-grid cell with an unadjusted t near two (mk_{63}'s sixth decile at $t = 2.34$, jump_5's seventh at 2.08, skew_{63}'s first at 2.01) loses significance once the family is priced, and the text treats them accordingly: the paper's claims rest on the cells that survive. Any cell that loses significance under the adjustment is identified as statistically indistinguishable from zero in the text wherever it is discussed. *Survivorship:* the holdout's observation filter conditions on names present for most of the era; the frontier comparison is relative (both models score the same names), so the filter tilts the universe toward survivors without by itself favoring either model in the ranking. The sharper version of the objection is selection on information unavailable at the start, and it too has a constructive answer: a point-in-time re-selection fixes the universe at the top 200 names by market capitalization as of the first quarter of 2000, applies no minimum-history filter, and keeps every name through its delisting date with the CRSP delisting return appended as the final observation. The point-in-time run is the strongest replication in the paper: on the 164 of 200 Q1-2000 names that clear the symmetric fitting floors (36 dropped; 79 delisting returns ridden to the end), the overall edge is +0.58% with per-date DM 8.96 over 2,692 dates, and the frozen design-era top bucket carries +2.94% at DM 6.6 (5.9% occupancy; the bottom bucket is +0.44%, DM 5.4 — in this era even calm cells lean positive, at one-seventh the top bucket's size). The edge is *larger* than in the survivor-tilted holdout: the observation filter turns out to have biased the reported edge downward, not upward. Names that later die are where shapes break; a universe that keeps them is kinder to the flexible engine and harder on the leakage-and-selection story.

5 The misspecification frontier

Conventions. Conventions are fixed once, here. (i) *Edges* are percentage reductions in pinball loss, so *larger is better* for the named model; a "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. (ii) *Ratios* are model-over-benchmark loss ratios, so *below 1 favors the model* and $1-$ratio is the edge ($0.973 =$ a 2.7% win; $1.043 =$ a 4.3% loss). (iii) *DM* is the Diebold–Mariano t-statistic on the loss difference. *Per-date DM* means, concretely: losses are first averaged across names within each date, $d_t = N_t^{-1}\sum_i(\ell_{it}^A - \ell_{it}^B)$ with the orientation stated at each use (benchmark minus engine in the engine's headline comparisons, so positive favors the engine), and the test is run on the resulting ∼1,100-date series with a Newey–West (lag-10) variance—the effective sample is the number of dates, never the number of asset-days. Reported p-values are *one-sided* in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%); t-statistics above 2.6 are significant under either convention, and marginal cases are flagged where they occur. A *tie* means the per-date DM cannot reject zero at the two-sided 10% level; the point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible the paper makes it as one: for the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin — a two-one-sided-tests conclusion, not a failure-to-reject. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic and a stationary bootstrap (mean block length 10 days; $B = 1{,}000$ replications in the main battery, $B = 800$ in the CAViaR extension), following Hansen et al. (2011). (iv) *Coverage* numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism); for *credible-interval* coverage, closer to nominal is better in both directions. (v) *Correlations* quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

5.1 The within-universe result

On 200 CRSP names (≈221.6k test asset-days), we bucket the per-day pinball edge of the amortized nonparametric model over GARCH-t into deciles of each misspecification signal. Table 4 shows the result that organizes the paper.

The edge is a top-decile phenomenon. It is large and significant where the fixed-t shape assumption is currently broken, and statistically indistinguishable from zero where it is not.

The score is model-relative, and a normalization makes that testable: raw trailing kurtosis mixes the heavy tail a correctly specified t_{ν_i} already implies with local departure from that law, so we convert each observation's mk_{63} into a percentile of the simulated null of 63-window sample kurtosis under the name's own fitted t_{ν_i} (population kurtosis does not exist below $\nu = 4$ and the median fitted ν is 3.9; the finite-window null is the only valid normalizer). The re-sorted frontier is unchanged: top decile +3.0% (DM 8.5) against +3.0% (DM 10.5) raw, deciles one through eight at zero, 82% top-decile overlap. The score reads departure from the fitted law,

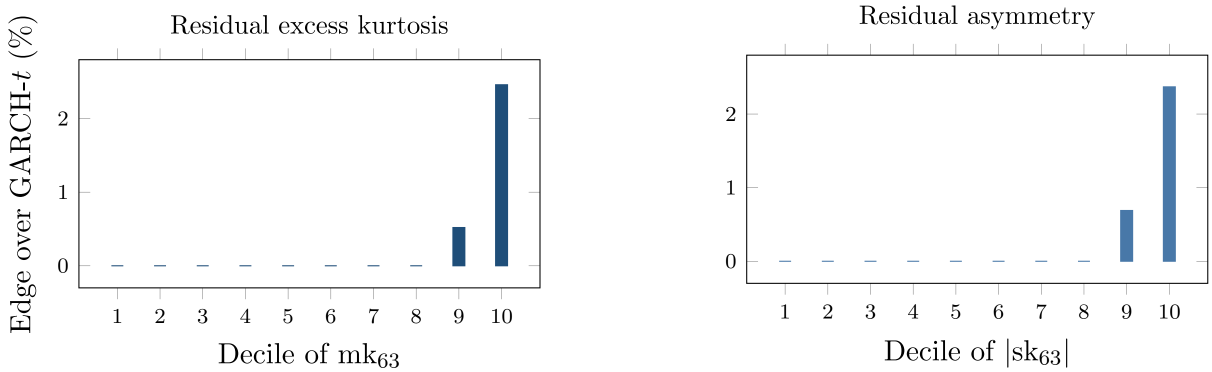

Figure 2: The misspecification frontier. Nonparametric pinball edge over GARCH-t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

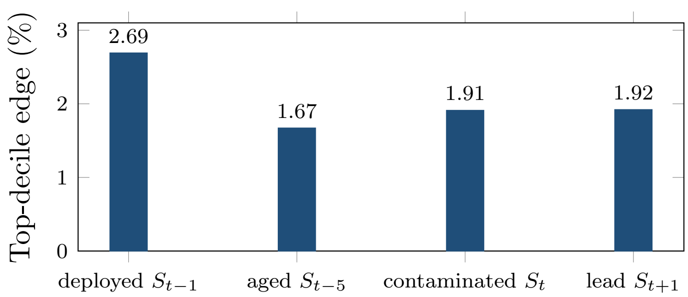

Figure 3: Timing diagnostics for the frontier. Top-decile edge under four alignments of the mk_{63} score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less — the opposite of what a look-ahead artifact would produce.

not raw tail mass.

Timing diagnostics. The score is computed from residuals through day $t-1$ and paired with the day-t loss; Figure 3 probes that alignment. The deployed lag gives the largest top-decile edge (+2.69%, DM 6.5); a five-day-old score retains +1.67% (DM 6.4). A deliberately contaminated same-day window, the alignment a look-ahead artifact would favor, yields *less* than the deployed lag (+1.91%; a one-day-ahead placebo gives +1.92%). Same-day sorting does not manufacture the edge.

An absorption test. If the score were merely an omitted variable, a parametric-adjacent fix should capture the edge without a pooled learner: per-name FHS with the residual window reweighted toward days whose mk_{63} resembles today's (Gaussian kernel). On the full panel the reweighted FHS is significantly *worse* than plain FHS overall (per-date DM -11.3) and in the top decile (pinball 0.541 against 0.444), while the pooled model beats plain FHS there by 2.7% (DM 8.8). The mechanism is effective-sample collapse: sharp univariate conditioning shrinks a name's usable history exactly when today is unusual. Pooling across names is what makes state-conditioning affordable — the amortized model borrows the shape information from every name's stressed days. The amortization of Section 3.3 is a precondition for the frontier result; a single name's history cannot support sharp state-conditioning on its own.

Robustness of the score. Table 5 double-sorts the edge on the score and on trailing volatility. Sorting test asset-days independently into quintiles of mk_{63} and of trailing volatility (rv_{63}): high volatility without a shape break carries no edge (top-volatility column, bottom four kurtosis quintiles: −0.57% to +0.10%), a shape break without an active regime carries almost none (top-kurtosis row, calmest quintile: −0.10%, DM 0.7), and the product state carries +2.33% with DM 6.4. The two ingredients interact. The score flags a currently wrong parametric shape; volatility scales the cost of the error. Second, a superior-predictive-ability test (Hansen, 2005) on the per-date loss panel against the full desk field (GARCH-t, FHS, EWMA, HS) returns $T^{\mathrm{SPA}} = 0$, consistent $p = 1.00$: no rival's advantage is even positive, and each is individually worse with $t \geq 5.8$. Third, the score needs no recalibration to stay useful: holding the design-era top-decile threshold fixed, the top-decile edge is positive in every test year (between +1.7% and +3.4%, 2020–2024), even though the cross-sectional 90th percentile of mk_{63} itself drifts upward in 2023–24. Fourth, the score is a slow state, so monitoring it costs little: a median name enters the top decile less than once a year (0.68 entries) and a median episode lasts 24 trading days.

The untouched-era holdout. Every design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze the entire pipeline (signals, features, hyperparameters, split rule, and test statistics), wrote down three predictions in advance (top-decile edge positive and significant; bottom decile indistinguishable from zero; a threshold-shaped decile profile), and then pulled CRSP daily returns for 2000–2013, an era no prior run in this project had touched. The frontier replicates (Figure 4): the top-decile edge is +0.89% with per-date DM 2.16 ($p = 0.016$; 1,465 dates), deciles one through nine sit between −0.18% and +0.28%, and the overall edge is +0.02%: the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. The holdout spans the 2008 crisis, so the replication is not an artifact of a calm era. One measurement choice in the replication deserves its own audit: decile boundaries above are within-sample ranks of the holdout test era (the standard conditional-sort convention), so the partition rule, though it never touches a forecast or a loss, is not a rule a desk could have applied on day one. The restatement a desk could have run: freeze the design era's numeric decile boundaries (the pooled design test sample's mk_{63} edges, top threshold 13.67) and apply them to the holdout as constants, so membership at (i, t) depends only on name i's own past 63 days and a number fixed before any holdout day was scored. The threshold shape survives the frozen rule (bins eight through ten carry +0.82%, +0.92%, +1.02% against −0.3–0.0% below), and the top-bin point edge is slightly *larger* than the within-sample sort's (+1.02% vs +0.89%), but the cell thins to 3.3% occupancy in this lower-kurtosis era and the per-date test loses power (DM 1.11, insensitive to Newey–West lags 5–44). A second real-time rule needs no design-era constant at all: an expanding pooled 90th-percentile threshold over strictly

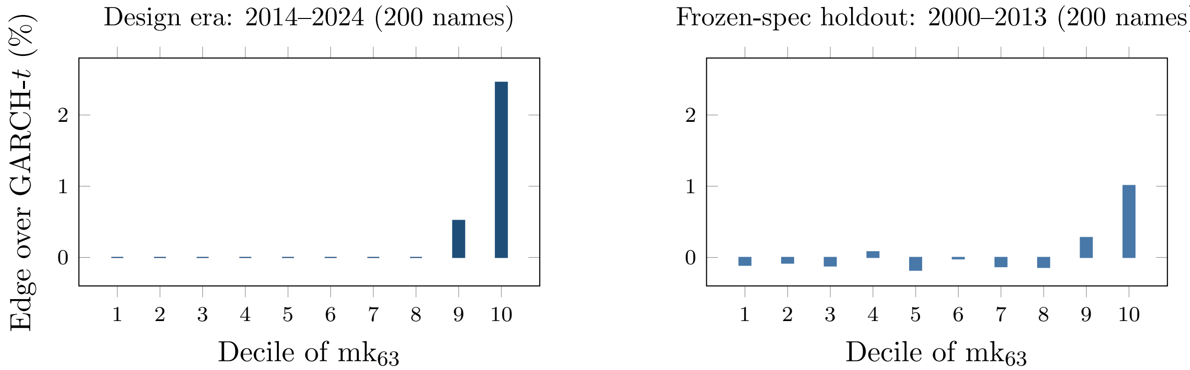

Figure 4: The frontier out of era. Left: the design-era decile profile (Table 4). Right: the same pipeline, frozen and run once on 2000–2013 CRSP data under the frozen specification. Bars are pooled per-observation decile means (top decile +1.01%); the inference statistic is the per-date aggregation of Section 3.7, +0.89% with DM 2.16 over 1,465 dates. All other deciles sit between −0.18% and +0.28%. The threshold shape survives an untouched era containing the 2008 crisis.

earlier dates (250-date burn-in) restores natural occupancy (9.5%) and recovers the sort's result with significance: +0.99%, DM 2.15, insensitive to Newey–West lags 5–44; the bottom bucket is −0.18% ($t = -1.4$). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge; the recursive rule is the version a desk could have run from day one (the frozen thresholds' own decisive test is the point-in-time universe of Section 4.2: +2.94%, DM 6.6).

The low overall signal–edge correlation (0.05–0.10) is the point, not a weakness: a threshold nonlinearity, summarized correctly by the top-decile Diebold–Mariano statistics.

5.2 The frontier across universes

Table 6 shows the same axis at work across markets. First, the only decisive *standalone* single-asset wins anywhere are the hyperinflation and capital-control currencies — Argentine and Egyptian pesos, 12–14% better with significance — whose residual kurtosis (∼2000–3300) sits two orders of magnitude above anything GARCH-t was designed to absorb. Second, the country gradient is mediated by residual kurtosis, not by the development label per se: frontier indices (kurt ∼10) sit between well-behaved developed indices (kurt ∼5) and hyperinflation FX on the same frontier. Third, the Korea-2026 negative control sharpens the mechanism: a dramatic *price* crash with circuit breakers is *not* residual misspecification — Korean residual kurtosis stays in the ordinary 4–13 range, and GARCH-t significantly wins all twenty-three assets even inside the crisis window. The frontier is about the residual shape, not about headline turbulence.

One adjustment prices the breadth of this search. The three exploratory universes contain 93 instrument-level comparisons; treated as one family, a Holm step-down at familywise 5% (Online Appendix Table OA.3) keeps eight numerical survivors, of which one (PJM power) is already reclassified by the audit below as a log-return artifact on near-zero prices, leaving seven: the three volatility indices, Baltic freight, wheat, USDEGP, and Kenya. Sri Lanka, Argentina,

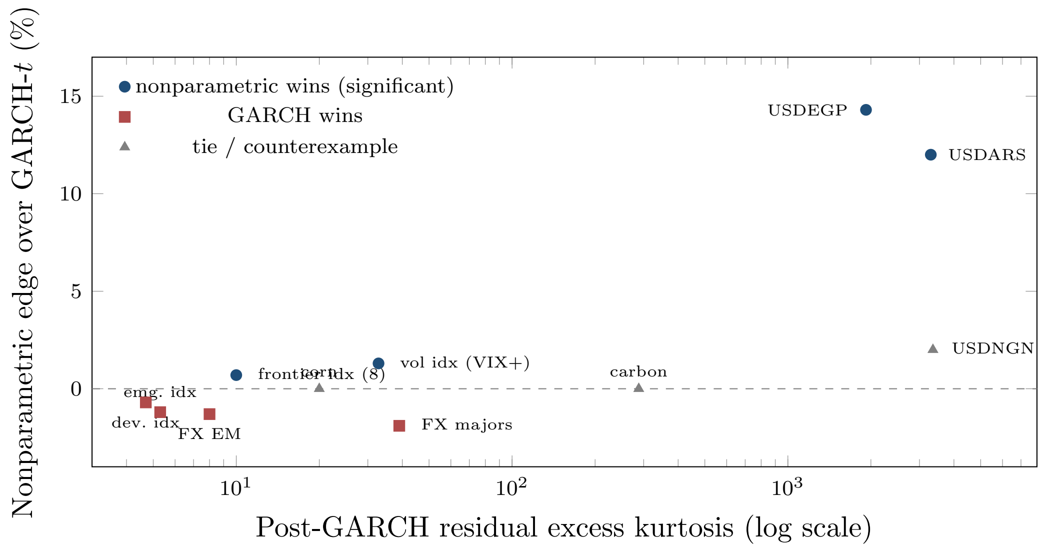

Figure 5: The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ∼2000–3300) carries the only decisive standalone wins; high kurtosis is necessary but not sufficient — carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

Pakistan, and Nigeria drop from the individually significant list. Instrument-level entries are therefore exploratory; the replication evidence this section rests on is pooled and structural: the crisis-tier FX comparison (DM 4.12, $p \approx 2 \times 10^{-5}$, surviving any adjustment) and the kurtosis–edge gradients across universes.

A frequency corollary. Daily crypto belongs to GARCH (zero nonparametric wins on five liquid coins). At *hourly* frequency the picture inverts: the neural IQN beats the full benchmark ladder, including filtered historical simulation, by ∼0.6% with per-date DM $p < 10^{-7}$. Sample density is itself a frontier dimension: more observations per unit of regime raise the resolvable residual-shape signal. A systematic intraday map of the frontier is pre-committed and blocked only on tick-data access.

5.3 Regime concentration and the futility of gating

In the amortized panel the edge concentrates in turbulence: the GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent (roughly fourfold) while both nonparametric estimators beat GARCH in *every* quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478). A natural inference is that one should *gate*: run GARCH on calm days and switch to the nonparametric model when the score is high. It fails, instructively. A classifier gate trained on the misspecification signals routes 90.5% of days to the nonparametric model and exactly matches always-nonparametric (0.3987 vs GARCH 0.4000); worse, its routing runs *backwards* across score deciles, because nonparametric wins on

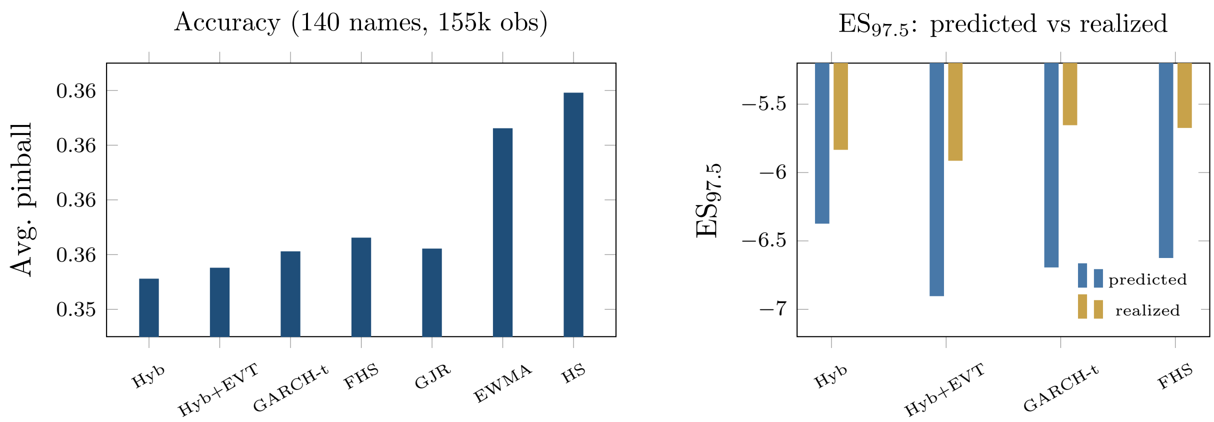

Figure 6: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 5.2–8.8 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted (exact tail integral) vs realized below each model's own VaR — a model-dependent diagnostic, not a ranking (every fat-tailed entrant's integral exceeds its own-tail mean); the joint FZ0 score is the ranking criterion.

high-score days are rare-but-large while a classifier targets frequency. A regression gate on the edge magnitude fixes the direction (routing flat ≈ 1.0, predicted-edge correlation $+0.16$) and still converges to always-nonparametric. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by *which day the tail event lands*, and that is unpredictable from prior-day state. The deployable conclusions: in the amortized setting, use the nonparametric model always (it does not underperform on calm days); use the residual-hybrid where a guaranteed $\geq$GARCH floor is wanted; and use the frontier score as a *monitor* (Section 7); day-level switching adds nothing. (Detection latency is small, median zero, and the edge is back-loaded within episodes, so the lag costs little even in the per-name setting.)

6 The FRTB-aligned forecasting battery

6.1 Head-to-head accuracy and significance

Table 7 reports the full battery on 140 CRSP names (155k obs, 1108 dates).

The stress-era replication. The battery's pre-committed re-run on the 2000–2013 holdout panel is complete. Splits follow Section 4, so each name's test window runs from roughly mid-2008 through 2013 — the crisis and its aftermath. The EVT-tailed engine passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t-statistics of 0.17 at 99% and 0.35 at 97.5% — consistent with nominal at both regulatory levels through the crisis window. The two models most common on desks do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH-t is also well calibrated in this era (94% pass rate), which is the frontier's own prediction — calibration parity is the norm, and the engine's added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative out of era (breach 2.21% at 97.5%, $t = -1.09$, not significant): the safety margin widens when the calibration era is calmer than the test era, and the direction of the error is the one regulators prefer.

The headline carries caveats. First, adding SAV-CAViaR, a model absent from banks' standard toolkit but the strongest semiparametric academic benchmark, produces a statistical tie: CAViaR 0.3461 vs hybrid 0.3460 (DM 0.59, $p = 0.28$; the CAViaR study's common sample differs slightly from Table 7's, which is why the hybrid's loss level is 0.3460 there against 0.3551 in the table; rankings, not levels, are comparable across the two runs); the 90% Model Confidence Set contains exactly these two. The hybrid is *co-best with CAViaR*, not uniquely dominant, and still significantly beats everything banks actually run. The tie is asymmetric in capability: SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history (no new listings). We do not hold pooled exception rejections against it: pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, ours included; the date-clustered restatement is the reading this paper trusts. The accuracy tie is between a single-level autoregression and one shared model carrying the whole deployment brief. Restating FHS the textbook way changes nothing: per-name train-constant and per-name rolling 500-day causal variants, run on the identical panel, lose to the hybrid by DM 5.4 and 5.8 (pinball 0.3562 and 0.3569 against 0.3550), bracketing the pooled variant the battery uses. Second, the neural IQN as residual model initially loses to trees badly (0.3650 vs 0.3551, DM 13.3) with a severely thin tail (breach$_{99}$ 3.2%); the tail-aware, monotone, conformally recalibrated version closes to a hair (0.3638 vs 0.3632, DM 3.03) with calibration repaired (breach$_{99}$ 0.92%). Trees remain the strongest tabular estimator; the tuned IQN is competitive. Third, the ES columns are a diagnostic, not a ranking. Under the exact tail integral (a numerical correction: the earlier three-node tail average understated predicted ES by 2–6%), every fat-tailed entrant's predicted ES exceeds the realized mean below its own VaR, by 9% (raw hybrid, rolling FHS) to 22% (pooled FHS), and the comparator conditions on each model's own breach set. The joint FZ0 re-scoring below is the ES ranking this paper stands on.

The joint $(\mathrm{VaR}, \mathrm{ES})$ score. Because ES is not elicitable alone but the pair $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the battery was re-checked under the FZ0 joint score (orientation validated by Monte Carlo against a known-truth t_5 before any ranking was read; first sandbox subsets of the panel, full-panel re-run pre-committed). On a 45-name large-cap subset, a Gaussian-innovation GARCH is significantly *worst* at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, fat tails are strictly necessary. Among the fat-tailed entrants — GARCH-t, FHS, and the hybrid+EVT — FZ0 differences on large caps are statistically indistinguishable ($|\mathrm{DM}| < 1.4$): on the well-behaved segment the EVT stage's distinct value is its exception-test compliance, a calibration property the FZ0 score does not reward. On a 40-name top-kurtosis subset the ordering flips and the hybrid+EVT attains the lowest FZ0 — the frontier pattern reappearing under a jointly consistent score — directionally but not yet significantly at this width (DM ≈ 0.9, $p \approx 0.17$).

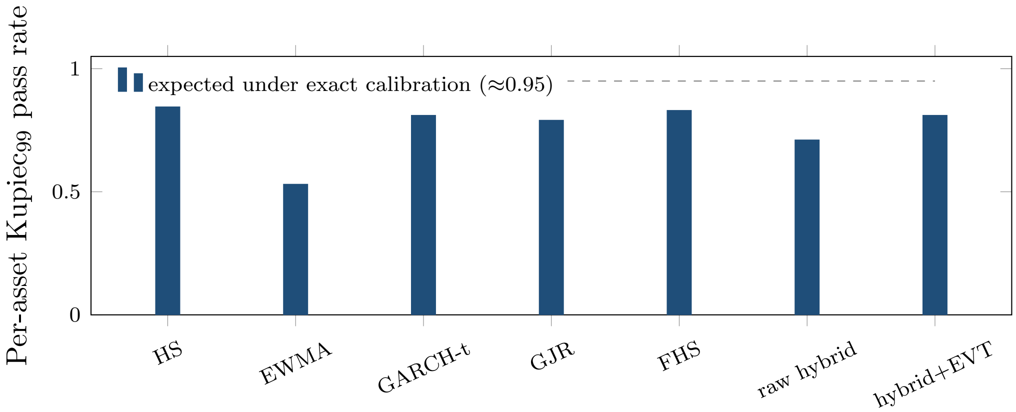

Figure 7: Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and the joint FZ0 score (the GJR bar predates the skew-t correction of Table 7's note; the corrected pooled row passes both 99% tests) (Table 7). Date-clustered tests in the table notes.

On the full 200-name panel (221.6k test asset-days), with ES from the fitted GPD tail and per-date DM inference throughout, the engine's *accuracy layer*, the EVT-tailed forecaster with no conformal shift, attains the lowest FZ0 at both regulatory levels: at 1% it beats GARCH-t (DM 4.0), FHS (DM 4.7), and a per-name score-driven one-factor GAS model estimated by FZ0 minimization in the style of Patton et al. (2019) (DM 9.3; a three-start re-estimate reproduces the 2.5% comparison at DM 5.9 but destabilizes at 1% for a few names; the benchmark is read as appendix-grade); at 2.5% it again wins (DM 4.4 over GARCH-t, 6.9 over GAS); the accuracy layer is thus a single unified configuration: lowest FZ0 at both levels *and* a pass on the aggregate date-clustered exception test at both levels, with per-name diagnostics showing residual cross-sectional calibration heterogeneity (81%/86% pass rates at 99%; Figure 7), with only the hypersensitive pooled 155k-observation Kupiec at 97.5% against it. The frontier concentrates the joint-score edge as it does the pinball edge: the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.6% against the 2.5% target) and gives back the FZ0 advantage there: the *static-overlay* variant loses to GARCH-t on the 2.5% joint score (DM -1.6, not significant at the two-sided 5% level), and the concession is largest exactly where the bands are widest: in the top misspecification decile the static shift's joint-score deficit reaches DM -6.7. The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative (wider bands, fewer breaches than nominal). A desk that prizes the conservative coverage margin keeps the shift; one scored on FZ0 efficiency runs the EVT tail alone, which passes the exception battery on its own (Figure 7). The concession is mostly *staleness*, not structure: the adaptive form of the shift (Gibbs and Candès, 2021), updated daily from the panel breach frequency, walks the margin back out of era (-0.35 to -0.10; breach 1.8% $\rightarrow$ 2.1%) and erases the overall concession: the adaptive

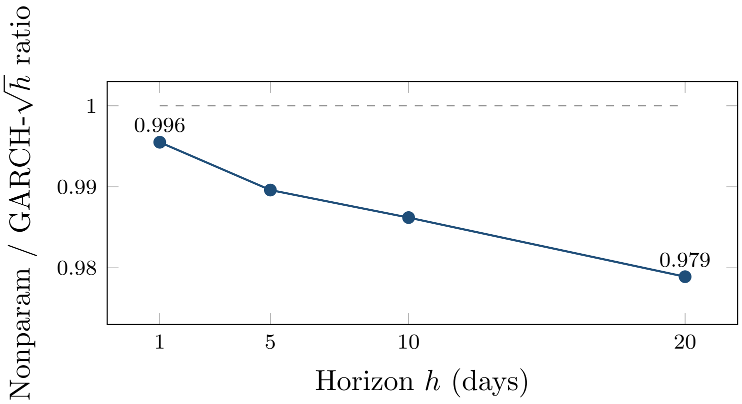

Figure 8: The edge grows with horizon. Direct-model/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (150-name design panel; training labels whose h-day window crosses the split are purged): fat tails compound while $\sqrt{h}$ scaling compounds the shape error — 0.45% at h=1 to 2.1% at h=20. The forecaster is the direct multi-day model of the main text's ten-day subsection, *not* the residual-hybrid engine, whose architecture is one-day. An earlier version of this figure showed larger ratios from an unpurged run; the purged rerun supersedes it.

variant ties GARCH-t on 2.5% FZ0 (DM -0.3) where the static loses (DM -2.0, same run). The top-decile cost survives any shift (-4.8 vs -6.0): the irreducible price of a margin where bands are widest. The static shift also buys per-name coverage badly (Kupiec$_{97.5}$ pass rate 47% against 72% unshifted, 69% adaptive; over-coverage fails from the other side). Every "lowest FZ0" headline refers to the accuracy layer; the static concession is a frozen margin's price, and the adaptive form removes the overall cost.

6.2 The ten-day horizon: a direct multi-day extension

The multi-day results in this subsection come from a *direct* amortized model — a pooled quantile learner trained on h-day cumulative returns with GARCH-derived features — and *not* from the residual-hybrid engine of Algorithm 1, whose residual architecture is one-day; the two are separate objects, and no ten-day claim in this paper attaches to the deployed engine. Training labels whose h-day window crosses the split are excluded, so no label contains test-era returns. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) the direct model's pinball edge over $\sqrt{h}$-scaled GARCH grows from $\sim$0.4% at $h = 1$ to $\sim$1.4% at $h = 10$, and holds in the 2020–2022 stress window (hybrid 1.248 < FHS 1.2605 < GARCH 1.262). The design's h-curve (Figure 8) rises from 0.45% at h=1 to 2.1% at h=20 under the same purge.

More consequentially for tail severity — with ten-day ES now the same exact tail integral as Section 6, and the same model-dependent own-tail caveat attached: every ten-day tail over-states its own-tail comparator, GARCH $\sqrt{h}$ most (predicted -21.1 vs realized -15.4) and the direct model least (-20.1 vs -17.3), with the best ten-day breach$_{97.5}$ (2.79% vs GARCH 3.30% against the 2.5% target). Fat tails compound with horizon; $\sqrt{h}$ Gaussian-style scaling compounds the shape error. A caveat: the temporal split places the whole test window in 2020–2024, and the pre-committed rerun on 2000–2013 shows the ten-day result is era-dependent. Out of era the two methods tie on accuracy (direct edge +0.49%, date-level DM 0.07, NW lag 10), and the calibration verdict reverses: $\sqrt{h}$-scaled GARCH-t ES is almost exactly calibrated through the crisis window (predicted -16.6 vs realized -16.6, exact integral, purged) while the direct model's tail under-states it (-15.9 vs -17.8) — the pooled ten-day tail, trained on the pre-crisis segment, did not extrapolate to 2008-scale ten-day losses. The ten-day tail-severity advantage is therefore a 2014–2024 statement, and a desk holding the engine at $h = 10$ through a 2008-type era should keep the parametric $\sqrt{h}$ number alongside as the conservative envelope. The frontier itself is confirmed on 2000–2013 (Section 5); this horizon result completes the pre-committed comparison with a mixed verdict, reported as such.

6.3 Cross-asset winners, with corrections

The clean calibrated cross-asset win is VIX (pinball 1.127 vs 1.137, DM 1.74, $p = 0.041$ one-sided, passes Christoffersen at both levels), a marginal result: it does not survive a two-sided convention, nor a multiplicity adjustment across the 43-instrument sweep, and is reported as suggestive rather than confirmatory. Credit (52% accuracy edge, DM 30.6) and Baltic freight (15%, DM 9.6) fail breach-independence (exceptions cluster) and are reported as accuracy edges with open calibration problems, not wins. (These are the FRTB-grade single-asset fits. Freight's magnitude agrees with the lean hybrid-vs-GARCH sweep of Table 6 (ratio 0.846 there); credit does not (the lean sweep shows only 4%, DM 2.3), because the two batteries pair different model variants against different benchmarks. Both are reported; the batteries are not interchangeable, and credit's larger figure should be read with that caveat in addition to its clustered exceptions.) An earlier electricity "win" was an artifact: log-returns on near-zero and negative power prices; refit on price *differences*, GARCH wins ERCOT (DM -2.55) and PJM (DM -2.73), and electricity is removed from the win list. The GPD tail also *hurts* some volatility series (VVIX, V2X, MOVE flip from win to tie/loss), motivating the selective-application rule of Section 3.2.

7 Applications and deployment

The production loop. *Nightly:* (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau \mid s_t)$ — one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and, where the desk elects the coverage overlay, the conformal shift c_τ (static or adaptive); (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\sigma}_{t+1}\,\widetilde{q}(\tau \mid s_t)$ at the desk's levels, with $\mathrm{ES}_{97.5}$ from the GPD closed form below the splice; (v) read the misspecification score (9) as the model-risk monitor — top decile means the parametric fallback's conditional average loss in that score region runs $\sim$2–3% higher. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* run the separate characteristics-only amortized forecaster until an own-history sample sufficient to initialize the per-name scale filter accrues; then attach this residual-hybrid pipeline. Total marginal cost per name per day: one filter update and one model evaluation.

Ex ante use of the score. Because the frontier score is a trailing window on GARCH-standardized residuals, it is available *ex ante*: a desk computes today's score from today's data and acts on it for tomorrow's risk number. What the evidence does *not* support is day-level model-switching. As Section 5.3 shows, the per-day oracle gap is unreachable (predicted-edge correlation +0.16), because which day a tail event lands is not forecastable from prior-day state, and a learned gate collapses to always-nonparametric. The edge is, in effect, negligible over the calm ∼90% of asset-days and concentrated (+2.7% pinball, DM 6.5) in the top misspecification decile and the most turbulent volatility regimes (an ∼fourfold gap, Section 5.3). A desk should therefore run the engine always and read the score as a monitor: on calm days the nonparametric/residual-hybrid family is statistically indistinguishable from the parametric benchmark (noninferior within the −0.25% margin where tested) and it wins in stress, so holding it is cheap and the stress-period edge accrues automatically. The one setting in which the score does drive a switch is the per-name/standalone model, where nonparametric quantiles underperform on calm days; there the score gates, and its short detection latency (median zero, mean two to four days) costs little because the edge is back-loaded within a misspecification episode.

Regulatory capital. The hybrid engine is an internal-models VaR/ES candidate that tops the accuracy table while passing the aggregate date-clustered exception test at both levels (per-name diagnostics show residual cross-sectional heterogeneity). This paper makes *no* capital-release claim: the predicted-vs-realized ES wedge conditions on each model's own breach set and doubles under the exact tail integral, so no defensible arithmetic converts it to capital. What survives audit is a mechanism stated with its limits: the ES-based component of an IMA charge scales with the ES forecast, but actual FRTB capital also involves liquidity-horizon adjustment, stressed-period ES on a reduced factor set, constrained aggregation, non-modellable risk factors, and desk-level eligibility through backtesting and P&L attribution — so no one-for-one mapping from a single-name ES forecast to desk capital exists, and none is claimed. Establishing genuine calibration differences requires a jointly valid standard, and the FZ0 comparison of Section 6 is the version of that comparison this paper stands on. The ten-day rerun on 2000–2013 shows $\sqrt{h}$-scaled GARCH almost exactly calibrated through the crisis era while the direct-learned tail under-states it: multi-day capital economics would be regime-dependent in the direction that matters most.

A misspecification monitor. The score is a live, per-asset monitor: an asset entering its top decile sits where the parametric model's conditional loss runs ∼2–3% higher, and the input is a trailing window on GARCH residuals, so surveillance is real-time and cheap.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant in this paper that can produce a conditional risk forecast for an asset with *no return history*: an IPO or newly listed ETF gets day-one quantiles from its characteristics and its first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names. (The day-one forecaster is the raw amortized model; the residual-hybrid engine of Section 3.2 attaches once a scale filter is estimable.) The practical numbers: the amortized forecast beats own-history benchmarks at *every* listing age, largest when young (∼6–10% of pinball in the first trading month, full-scale panel), and below our 750-observation fitting floor a per-name GARCH is not reliably estimable in this panel. Section 3.3's ablation shows how the machine works as history accrues: characteristics carry the forecast at birth, and the name's own recent realized volatility takes over as it ages, because the model learned one universal state-to-quantile mapping rather than per-name parameters.

Trading implications. The universes where the nonparametric tail wins — crisis FX, volatility indices, credit spreads, frontier equities — are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physical-measure tail forecast does not by itself make bought crash insurance profitable, because the crash risk premium can survive conditioning on a correct forecast. The deployable side is the other one — *warehousing* (selling) the premium — with the model supplying tail control rather than alpha on the mean: model-timed overlays improve tail outcomes, not average returns, consistent with the tail being the only place the model knows something a fixed-shape filter does not. The contribution to a premium-harvesting book is risk management (when to size down or step aside ahead of a genuine crash), not signal generation.

Scenario generation. Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

8 Conclusion and limitations

A fixed-shape parametric filter is hard to beat where its assumption holds: most assets on most days. The paper's answer to "when and why beat it" is a measurable frontier: local violation of the residual-shape assumption, at whose top decile the distribution-free family wins decisively. The engine that captures this beats the market-risk benchmarks banks commonly run on FRTB-aligned diagnostics; the full IMA approval machinery is not claimed.

Limitations. Gradient-boosted trees remain the strongest tabular estimator; the IQN earns its place through sampling rather than point accuracy. The score is necessary-not-sufficient (carbon and corn tie at high kurtosis; Baltic Dry wins at moderate). Credit and freight show accuracy edges but clustered exceptions, unresolved. The 2000–2013 holdout and its battery re-run cover the frontier's 2008 exposure (Section 6); the intraday extension awaits tick data. The per-day oracle gap of Section 5.3 disciplines model-switching: day-level advantage is mostly realization noise.

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Data and code availability

A public replication package reproducing every WRDS-based table and figure in this paper (the Bloomberg-based exhibits are preserved as-run with their derived series — terminal access ended in mid-2026) — estimation code for all stages of the engine ((2)–(5)), the misspecification score (9), the FRTB battery, the generative-posterior and SBC routines, and the exact walk-forward, seeds, and hyperparameter grids referenced in the text — is public at `github.com/sean-qin-usa/downside-risk-paper` (the repository README maps each table to its canonical script and result artifact; the strict-calibration audit is `code/job_fz_strict_calibration.py` → `results/fz_strict_calibration_results.json`); a citable archival snapshot (Zenodo DOI) will be minted for the accepted version. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author's institutional licenses and cannot be redistributed; the package includes the code to rebuild every panel from those sources, the derived non-proprietary series, and a synthetic example dataset (`toy_example.py`) that runs the full pipeline end-to-end without licensed data. The holdout's frozen specification, including its written-in-advance predictions, is the header of `code/job_wrds_holdout.py`, line-for-line diffable against the design-era pipeline in the same folder. Two numerical corrections are canonical in the package: the GJR-skew-t inverse CDF (history preserved in `code/frtb_bench.py`) and the exact-tail-integral ES. Table 7 is generated by `code/frtb_table.py` → `results/frtb_table_results.json`; the ten-day figures by `code/frtb_stress_exact.py` → `results/stress_es_results.json`; pre-correction artifacts are preserved in the repository history, not in the working tree. The analyses deferred in earlier drafts and since completed ship with their result files, and the remaining open items (Section 8) are tracked in the repository.

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A Proofs

The four formal results of Section 3.4 are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]}(q-z)\,dF(z) \;+\; \tau\int_{(q,\infty)}(z-q)\,dF(z)$$

is finite for every q, and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed z, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q')-\rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau,1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z<q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z<q\} - \tau = F(q^-) - \tau,$$

which equals $F(q)-\tau$ for all but countably many q. Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u)-\tau\big)\,du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity*. For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u)-\tau\big)\,du = \int_{q}^{q_F} \big(\tau - F(u)\big)\,du \;\geq\; 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q. Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q, then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u-q_F| \leq |F(u)-\tau| \leq M|u-q_F|$, and integrating over the interval yields $\frac{m}{2}(q-q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q-q_F)^2$. □

Proof of Proposition 1 (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k-th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \ldots, e_n\}$ exists; when $k = n+1$ (i.e. τ within $1/(n+1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n+1$ errors $e_1, \ldots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$) are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n+1$ values is uniformly distributed on $\{1, \ldots, n+1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\left(z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\right) = \Pr(R \le k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \le \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n+1$ gives $\tau \le k/(n+1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1}, because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. □

Proof of Proposition 2 (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu\sqrt{(\nu-2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1+o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1+o(1))$ with $c_\nu = \tilde{c}_\nu((\nu-2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1+o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1+o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1+o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}}\,(1+o(1)) = \text{const}\cdot\tau^{-(1/\nu_1 - 1/\nu_2)}(1+o(1)) \;\longrightarrow\; \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of τ^*; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in Remark 1.) Under local truth t_{ν_1}, applying (8) with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. □

Proof of Lemma 2 of the paper (generative sampler)

Let $\widetilde{q}(\cdot \mid s)$ be nondecreasing and left-continuous on $(0,1)$ and $\tau \sim U(0,1)$, and set $Z^* = \widetilde{q}(\tau \mid s)$ and $G(x) = \sup\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ (with $\sup\emptyset = 0$). Monotonicity makes $\{t : \widetilde{q}(t \mid s) \le x\}$ an interval with left endpoint 0, and left-continuity ensures the supremum is attained when the set is nonempty, so the set equals $(0, G(x)]$; hence $\{Z^* \le x\} = \{\tau \le G(x)\}$ and $\Pr(Z^* \le x) = G(x)$, i.e. Z^* has CDF G. Because G so defined is the upper generalized inverse of $\widetilde{q}$, and a left-continuous nondecreasing $\widetilde{q}$ is in turn the lower generalized inverse of G, the quantile function of Z^* is exactly $\widetilde{q}(\cdot \mid s)$. When $\widetilde{q}$ is an estimated curve, the argument applies verbatim to the estimate: the sampler is exact for the *fitted* conditional law, which is the claim. □

Table 1: Summary of results and scope conditions. Edges are percentage pinball reductions unless noted; DM statistics are per-date (Section 5 conventions note), one-sided. Panel A: the deployed amortized residual-hybrid engine (worst case on the accuracy yardstick: no detectable loss; calm-cell 90% lower confidence bounds sit above −0.25%). A separate block reports the direct multi-day extension, which is not the engine. Panel B: boundary evidence from configurations not deployed, which motivates the design.

Panel A: the engine Result	Magnitude	Significance
vs GARCH-t / EWMA / HS, h=1 (avg)	+0.3 / +1.5 / +1.9%	DM 4.4–8.3
Top misspecification decile	+2.7%	DM 6.5
Hyperinflation FX (ARS, EGP)	+12–14%	DM 2.9–3.8
Frontier equity indices (4 of 8)	∼+1–3%	DM 2.1–3.5
2000–2013 frozen-spec holdout (untouched era)	top decile +0.9%	DM 2.2 (sort); recursive real-time rule +1.0%, DM 2.15; frozen-constant restatement in text
Point-in-time universe (Q1-2000 top 200, delistings kept)	top bucket +2.9%; overall +0.6%	DM 6.6 / 9.0, frozen real-time thresholds
Strict calendar splits; annual-refit walk-forward	top decile +2.5% under all three	DM 6.2 (2020 cut); 3.1 (overall, pre-GFC cut); 6.1 (walk-forward)
Familywise error over the 3 × 10 signal-decile grid	9 cells survive 5% FWER	Romano–Wolf step-down, incl. all three top deciles
Exception battery through the 2008 crisis	Kupiec 94% / Christof. 95% at 99%	date-clustered t 0.17 at 99%; at 97.5%, 0.35 (EVT tail) and −1.09 (static overlay, conservative); HS, EWMA fail
Joint (VaR, ES) FZ0 score, full panel	accuracy layer lowest at both levels	DM 4.0–4.7 vs GARCH-t, FHS; 6.9–9.3 vs GAS; static shift concedes 2.5% (DM −1.6), adaptive shift erases it (tie; top-decile cost remains)
Cold start / young listings	+6–10%	full-scale panel; no per-asset model applies
Calm-quintile edge	≈ 0 (point estimates +0.0–+0.2%)	bottom-decile 90% CIs lie above −0.25%; no detectable loss
CAViaR (strongest academic rival)	tie (co-best in MCS)	DM 0.59
Extension (a separate direct multi-day model, not the engine)		
Ten-day horizon vs $\sqrt{h}$-scaling	+1.4% (2014–24)	holds in 2020–22 stress; 2000–13 rerun: tie, DM 0.07 (era-dependent)
Panel B: boundary evidence from configurations not deployed		
Bare standalone neural net, single names	−4.5 to −6.2%	DM +12 to +18; motivates the hybrid design
Standalone nonparametric, calm single assets	GARCH as good or better	pooled DM −9.5 to −13.7; the shape edge needs misspecification
Korea 2026 crisis, 23 assets (negative control)	GARCH wins all	DM −2.5 to −5; a price crash is not a shape break
mk_{63}-reweighted FHS (reweighting probe)	worse than plain FHS	DM −11 overall and top decile; sharp univariate conditioning collapses the sample
Baltic freight; IG credit[a]	+15%; +52%	DM 9.6; 30.6; accuracy only, calibration open

[a] Credit and freight fail breach-independence (exceptions cluster); their entries are accuracy results only. The credit magnitude is from the FRTB-grade single-asset battery; the smaller 4% figure in Table 6 is from the separate lean hybrid-vs-GARCH sweep — different battery, both reported.

Table 2: Model selection by task and regime. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	§3.3
Single asset, calm (∼90% of asset-days)	GARCH-t	§5
Single asset, top misspecification decile	Nonparametric (+2.7%, DM 6.5)	§5
Amortized cross-name panel, any regime	Nonparametric always; no gate	§5.3
Regulatory ES with a ≥GARCH floor	Residual-hybrid + EVT + conformal	§6

Table 3: Sample flow across evaluation panels.

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map; engine development
Untouched-era holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	≈155,000	per-asset backtesting battery
Cross-asset sweep	Bloomberg	through 2026	43	[a]	frontier breadth
Country indices	Bloomberg	through 2026	26	[a]	frontier breadth

[a] Frozen exhibits; per-instrument windows vary and are documented instrument-by-instrument in the repository.

Table 4: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing residual-shape score (200 CRSP names, 221.6k test asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\lvert\mathrm{sk}_{63}\rvert$ (asymmetry)	jump_5
1–8 (bulk)	≈ 0	≈ 0	mixed, ≈ 0
9	+0.52	+0.69	—
10 (top)	**+2.46**	**+2.37**	+0.90
Overall		+0.44	

Per-date Diebold–Mariano: top mk_{63} decile edge +2.71%, DM 6.54 ($p \approx 0$); bottom decile +0.19%, DM 1.41 ($p = 0.079$, not significant); top two deciles pooled +1.65%, DM 6.74; overall +0.44%, DM 5.67. Top-decile signal means: kurtosis ≈ 25, |skew| ≈ 4. Three top-decile conventions appear in this section and differ by construction: pooled asset-day weighting gives +2.46 (the table cells), equal-date weighting gives +2.71 (the DM convention), and the timing-diagnostics run of Figure 3 reports +2.69 under its deployed one-day lag.

Table 5: Double sort: the edge needs both a shape break and an active regime. Cells are the nonparametric-over-GARCH-t pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

	rv_{63} Q1 (calm)	Q2	Q3	Q4	Q5 (active)
mk_{63} Q1 (low)	+0.24	+0.34	+0.10	+0.01	−0.14
Q2	+0.16	+0.20	+0.22	+0.09	−0.57
Q3	−0.02	+0.19	+0.25	+0.44	+0.04
Q4	+0.08	+0.15	+0.35	+0.23	+0.10
Q5 (high)	−0.10	+0.44	+0.68	+0.61	+**2.33**

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4. The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

Table 6: The frontier across universes. Ratio = nonparam/GARCH-t pinball (< 1: nonparametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM −13.7
	EM (10)	1.013	8	GARCH, pooled DM −9.5
	crisis (7)	0.973	∼1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86, $p = .002$
	USDEGP	0.857	1920	DM 3.80, $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins[a]
Cross-asset (43)	vol indices	∼0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
13/43 significant wins overall; corr(log kurt, edge) = 0.43				
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM −2.5 to −5

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06); exploratory — only Kenya survives the 93-comparison familywise adjustment (Online Appendix Table OA.3). Cross-index corr(log residual kurtosis, edge) = 0.53. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating necessity-not-sufficiency.

Table 7: FRTB battery: average pinball loss (twelve quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	Breach_{99}	Kupiec_{99} p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	−6.37 / −5.83	1.09%	0.00
+ EVT tail	0.3555	−6.90 / −5.91	0.91%	0.00
GARCH(1,1)-t	0.3561	−6.69 / −5.65	1.06%	0.026
GJR-GARCH-skew-t	0.3562	−6.58 / −5.84	1.05%	0.062
GARCH-FHS (per-name)	0.3562	−6.61 / −5.84	1.11%	0.00
GARCH-FHS (pooled)	0.3566	−6.90 / −5.67	0.99%	0.79
GARCH-FHS (rolling 500d)	0.3569	−6.53 / −5.98	1.09%	0.00
EWMA (RiskMetrics)	0.3606	−5.52 / −5.63	1.63%	0.00
Historical simulation	0.3619	−7.21 / −6.86	0.84%	0.00

One run, one artifact: every column is generated from `frtb_table_results.json` on a single common sample (155k asset-days; the rolling-FHS burn-in trims all models identically). Predicted ES is the exact tail integral $\alpha^{-1}\int_0^\alpha Q(u)\,du$: closed forms for the t and normal entrants, 200-node integration of the Hansen skew-t inverse CDF, the GPD closed form for the EVT tail, and a matched 20-node midpoint rule for the empirical quantile models. An earlier draft approximated predicted ES by a three-node tail-quantile average, understating it by 2–6%; the exact integral is why these levels differ from prior versions. Diebold–Mariano vs the hybrid: GARCH-t 5.22, GJR-skew-t 5.18, FHS pooled 6.43, per-name 5.42, rolling 5.88, HS 7.54, EWMA 8.79 (all $p \approx 0$); the EVT variant concedes 0.0004 pinball to the raw hybrid (DM 6.72) as the price of its tail; the 90% MCS contains the raw hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a *diagnostic*, not a ranking: "realized" is the mean return below each model's *own* VaR, so the conditioning set is model-dependent, and under the exact integral every fat-tailed entrant exceeds its own-tail comparator by 9–22%. The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew-t row: corrected Hansen inverse CDF with the fitted η, λ (median 4.2, −0.015; validated against the symmetric-t limit and `arch`'s own distribution) — the original implementation silently reduced to a symmetric $t(8)$; defect and correction are disclosed in the availability section.

Exception columns are pooled panel diagnostics: at 155k observations a breach-rate deviation of ±0.1 points rejects in either direction (the EVT variant's 0.91% sits on the conservative side; pooled FHS happens to land at 0.99%), so pooled p-values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement (Figure 7): per asset the EVT-tailed engine passes Kupiec at 99% for 81% of names and Christoffersen for 86%, and the date-clustered test passes at both levels ($t = 1.01$ at 99%, 1.86 at 97.5%) while the raw hybrid is rejected ($t = 4.5$) — the tail stages carry the regulatory calibration, and the 2008-era battery re-run confirms it out of era. At 97.5% the split-conformal recalibration is the stage that repairs pooled Kupiec (GBM-recal $p = 0.24$).